

PROJECT UPDATE

CSE440.1 ↘

PRESENTED BY – GROUP 10

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Project Overview

Overview:

Creating an intelligent racing game where AI agents learn to navigate tracks using reinforcement learning for autonomous driving simulation.

Objectives:

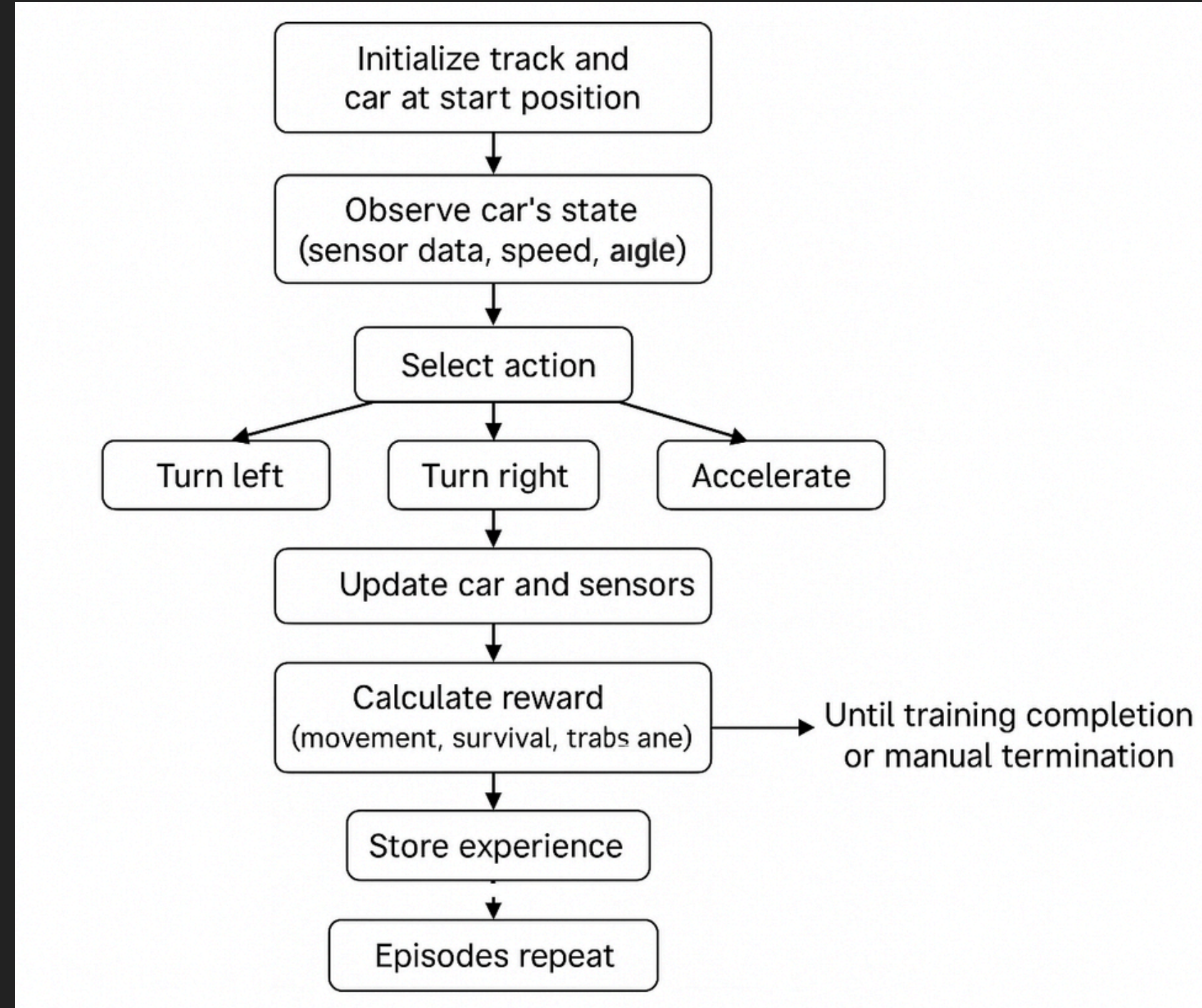
- Train an AI agent to drive a car on a racetrack using reinforcement learning
- Learn through trial-and-error to stay on track
- Optimize lap completion time
- Avoid crashes and penalties
- Demonstrate continuous improvement over training episodes



What We Built:

- The system is modular: the Game manages episodes, the Car handles movement and sensors, the Track defines boundaries, and the Agent makes decisions using a DQN.
- The car observes its environment using 5 front-facing sensors that detect wall distances.
- It receives a state vector of 7 values: sensor data, speed, and angle — all normalized for the neural network.
- Based on this, the agent chooses one of three actions: turn left, turn right, or accelerate.

System Overview & Architecture

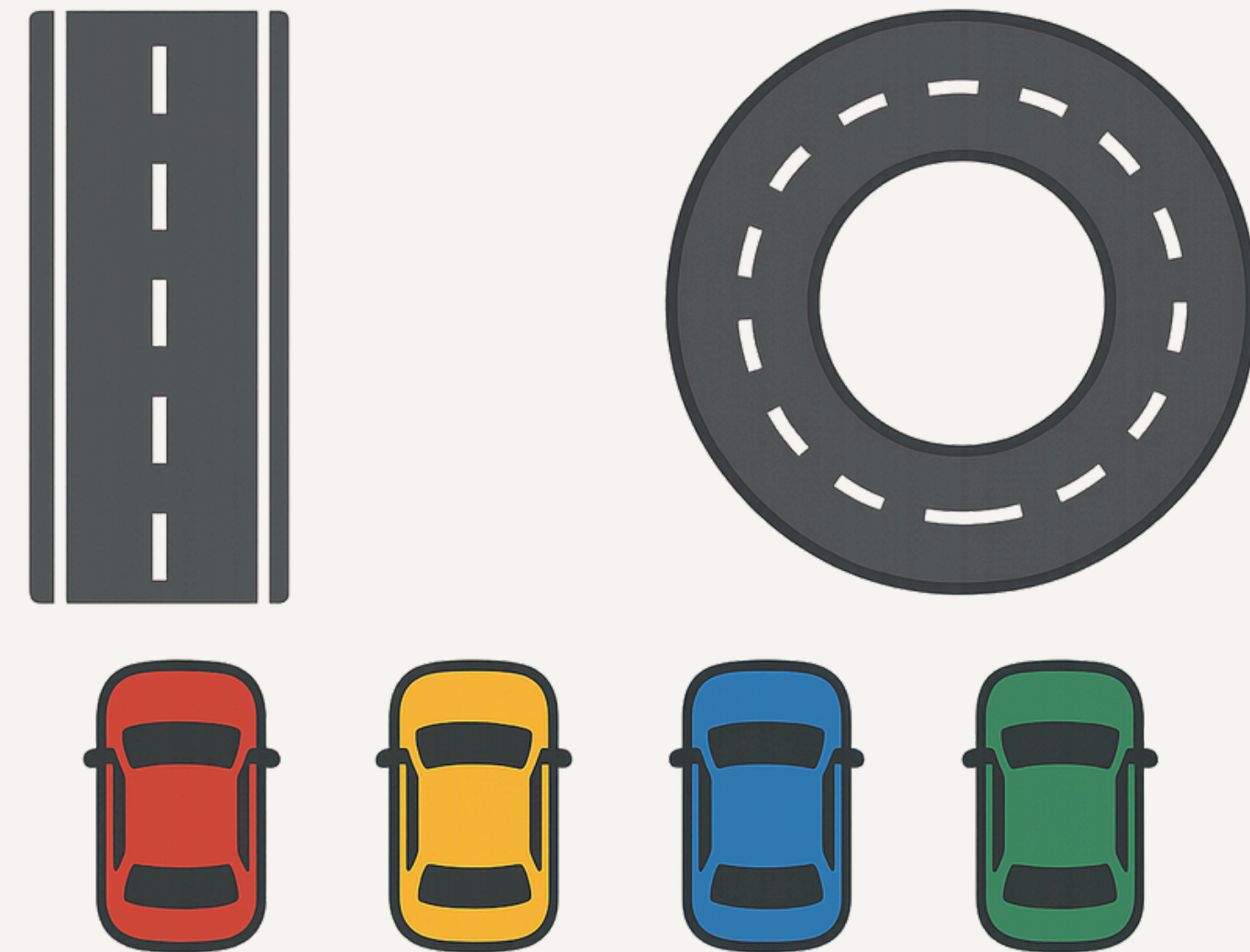


Reinforcement Learning & Training

- We used a Deep Q-Network to train the agent. It learns by trial and error, maximizing cumulative rewards.
- The reward function gives points for staying on track, moving forward, and surviving — and penalizes crashes heavily.
- To stabilize training, we used experience replay and a target network, which are key DQN techniques.
- The agent starts with random actions (exploration) and gradually shifts to smart decisions (exploitation) as epsilon (ϵ) decreases.

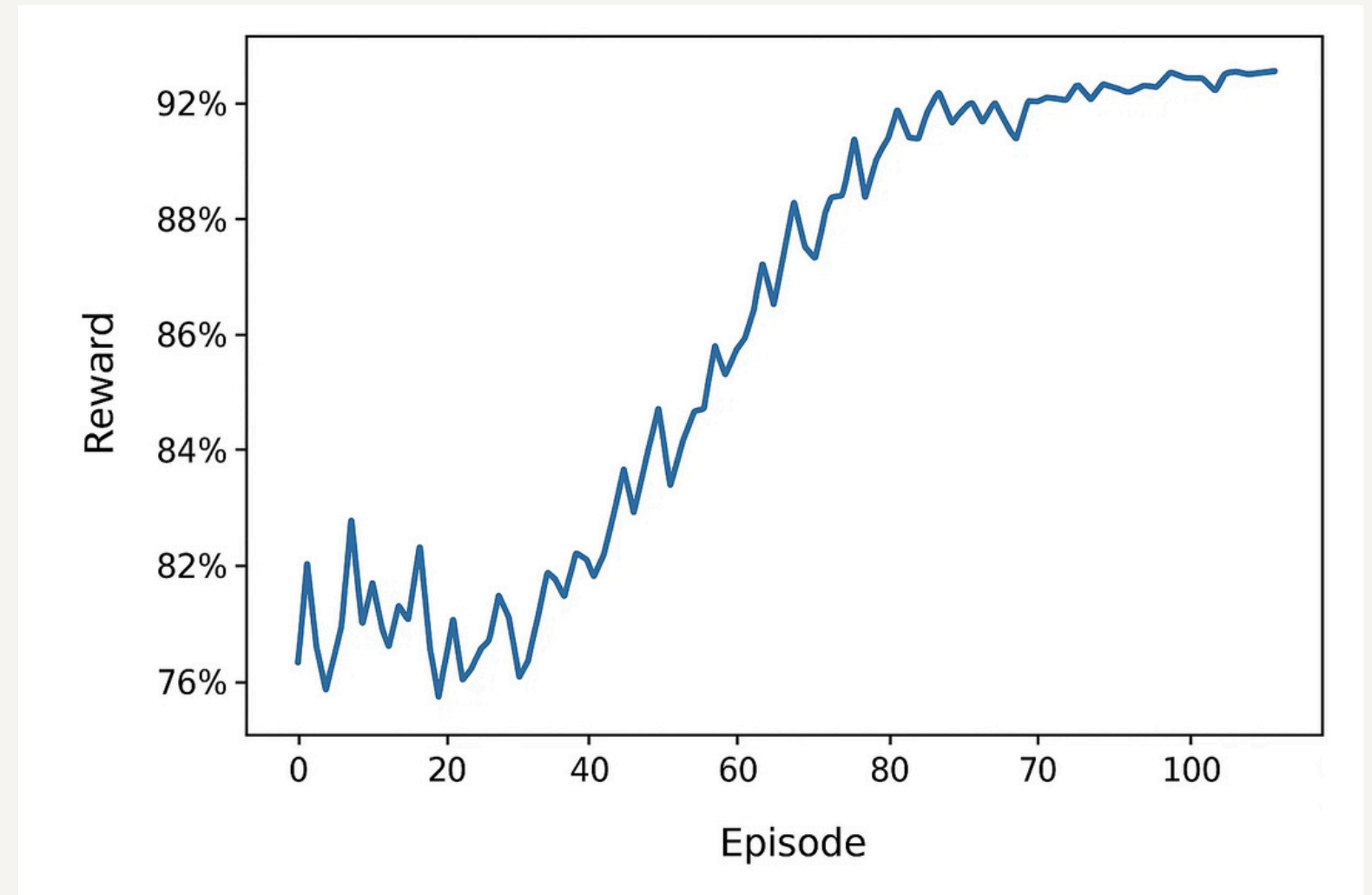
Game Mechanics & Collision Detection ↘

- The car moves using trigonometry: its position updates based on angle and speed, with friction applied.
- It stays within screen bounds using position clamping — this prevents it from disappearing.
- The track is a rectangular loop: the car must stay between the outer and inner boundaries.
- We implemented precise line intersection math to calculate sensor distances to walls.
- If the car goes inside the inner rectangle or outside the outer one, it crashes and the episode ends.



Training Progress and Outcomes ↘

- After training for several hundred episodes, the car learns to stay on track and drive smoothly.
- We track metrics like total reward, steps survived, and distance traveled — all of which improve over time.
- In testing mode, the trained agent performs consistently, with high scores and long survival times.



Strengths, Limitations & Future Work ↘

strengths

- GPU acceleration
- Clean code structure
- Responsive UI with early exit options.

limitations

- The track is basic
- Actions are limited
- The agent may get stuck in loops.

future steps

- Curved and more complex tracks
- Braking
- 360° sensors
- Multi-agent racing.

Thank You!

